R1

sid	bid	day
22	101	10/10/96
58	103	11/12/96

B1

bid	bname	color
101	tiger	red
103	lion	green
105	hero	blue

S1

sid	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S2

sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

Q1: $\pi_{sid,sname,age}(\sigma_{rating<10}(S_2))$

Q2: $\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

Q3: $\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

Q4: $S_2/\left(\pi_{sid,age}(S_2)/\pi_{sid}(\sigma_{bid=103}(R_1))\right)$

Q5: $(S_2 \bowtie_* R_1) \underline{\cup} (B_1 *\bowtie_* R_1)$

Q6: $\pi_{sname,rating,age,bname,color}\left(((S_1 \cup S_2) \bowtie_{sid} R_1) \bowtie_{bid} \sigma_{color=red}(B_1)\right)$

Q7: $\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$

Q8：Conver the query of Q1 to DRC formula and TRC formula.

Q9：Conver the query of Q2 to DRC formula and TRC formula.

Q10：Conver the query of Q3 to DRC formula and TRC formula.

Q11: What's data model? What's data schema?

Q12: Students(sid: string, name: string, ID number: string, cid: string)

Courses(cid: string, cname:string)

List all super keys, candidate keys, primary keys, foreign keys of table Students.

Q13: What are the relational operations on the relational data model? What are their differences and similarities?

Answer

Q1: $\pi_{sid,sname,age}(\sigma_{rating<10}(S_2))$

第一步：$\pi_{sid,sname,age}(\sigma_{rating<10}(S_2))$

“在 S2 表中选择，条件：排名小于 10”

sid	sname	rating	age
28	yuppy	9	35
31	lubber	8	55.5
44	Guppy	5	35
~~58~~	~~rusty~~	~~10~~	~~35~~

第二步：$\pi_{sid,sname,age}(\sigma_{rating<10}(S_2))$

“投影 保留字段：sid, sname, age”

sid	sname	rating	age
28	yuppy	~~9~~	35
31	lubber	~~8~~	55.5
44	Guppy	~~5~~	35

最终答案:

sid	sname	age
28	yuppy	35
31	lubber	55.5
44	Guppy	35

Q2: $\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

第一步：$\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

2 个操作 笛卡尔乘积+选择（条件：S1.sid>R1.sid）

S_1.sid	sname	rating	age	R_1.sid	bid	day
~~22~~	~~dustin~~	~~7~~	~~45~~	~~22~~	~~101~~	~~10/10/96~~
~~22~~	~~dustin~~	~~7~~	~~45~~	~~58~~	~~103~~	~~11/12/96~~
31	lubber	8	55.5	22	101	10/10/96
~~31~~	~~lubber~~	~~8~~	~~55.5~~	~~58~~	~~103~~	~~11/12/96~~
58	rusty	10	35.0	22	101	10/10/96
~~58~~	~~rusty~~	~~10~~	~~35.0~~	~~58~~	~~103~~	~~11/12/96~~

第二步：$\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

投影，保留字段 sname，rating:

~~S_1.sid~~	sname	rating	~~age~~	~~R_1.sid~~	~~bid~~	~~day~~
~~31~~	lubber	8	~~55.5~~	~~22~~	~~101~~	~~10/10/96~~
~~58~~	rusty	10	~~35.0~~	~~22~~	~~101~~	~~10/10/96~~

最终答案:

sname	rating
lubber	8
rusty	10

Q3: $\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

第一步：$\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

$\sigma_{bid}(R_1 \times B_1)$=$R_1 \bowtie_{bid} B_1 \neq \sigma_{R_1.bid=B_1.bid}(R_1 \times B_1)$:

两个表格笛卡尔乘积后按条件选择（等价于带条件的连接操作）

等值连接，条件：两个表格 bid 相等，但结果中的 bid 字段保留一个

sid	bid	day	bname	color
22	101	10/10/96	tiger	red
58	103	11/12/96	lion	green

第二步：$\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

投影，保留字段 sid,bname:

sid	~~bid~~	~~day~~	bname	~~color~~
22	~~101~~	~~10/10/96~~	tiger	~~red~~
58	~~103~~	~~11/12/96~~	lion	~~green~~

最终答案:

sid	bname
22	tiger
58	lion

Q4: $S_2/(\pi_{sid,age}(S_2)/\pi_{sid}(\pi_{bid=103}(R_1)))$

第一步：$S_2/(\pi_{sid,age}(S_2)/\pi_{sid}(\sigma_{bid=103}(R_1)))$

先分别投影,然后做除法

$\pi_{sid,age}(S_2)$ $\pi_{sid}(\sigma_{bid=103}(R_1))$ 结果

sid	age
28	35.0
31	55.0
44	35.0
58	35.0

除 以

sid
58

等于

age
35

第二步：$S_2/(\pi_{sid,age}(S_2)/\pi_{sid}(\pi_{bid=103}(R_1)))$

做除法

sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.0
44	guppy	5	35.0
58	rusty	10	35.0

除 以

age
35

等于

最终答案：

sid	sname	rating
28	yuppy	9
44	guppy	5
58	rusty	10

Q5: $(S_2 \bowtie * R_1) \cup (B_1 \bowtie * R_1)$

第一步： $(S_2 \bowtie * R_1) \cup (B_1 \bowtie * R_1)$:

基于 sid 右外连接，只保留R_1中所有的元组，无法匹配的内容 null

sid	sname	rating	age	bid	day
22	null	null	null	101	10/10/96
58	rusty	10	35	103	11/12/96

第二步： $(S_2 \bowtie * R_1) \cup (B_1 \bowtie * R_1)$:

基于 bid 全连接，保留R_1和B_1中所有的元组，无法匹配的内容 null

sid	bid	day	bname	color
22	101	10/10/96	tiger	red
58	103	11/12/96	lion	green
null	105	null	hero	blue

第三步： $(S_2 \bowtie * R_1) \cup (B_1 \bowtie * R_1)$:

外并：字段是两个表格的字段并集，不存在值填 full

sid	sname	rating	age	bid	day	bname	color
22	null	null	null	101	10/10/96	null	null
58	rusty	10	35	103	11/12/96	null	null
22	null	null	null	101	10/10/96	tiger	red
58	null	null	null	103	11/12/96	lion	green
null	null	null	null	105	null	hero	blue

Q6: $\pi_{sname,rating,age,bname,color}\left(\left((S_1 \cup S_2) \bowtie_{sid} R_1\right) \bowtie_{bid} \sigma_{color=red}(B_1)\right)$

第一步：

$\pi_{sname,rating,age,bname,color}\left(\left((S_1 \cup S_2) \bowtie_{sid} R_1\right) \bowtie_{bid} \sigma_{color=red}(B_1)\right)$

集合并集，重复项去除

sid	sname	rating	age
22	dustin	7	45
31	lubber	8	55.5
58	rusty	10	35
28	yuppy	9	35
44	guppy	5	35
~~58~~	~~rusty~~	~~10~~	~~35~~

第二步：

$\pi_{sname,rating,age,bname,color}\left(\left((S_1 \cup S_2) \bowtie_{sid} R_1\right) \bowtie_{bid} \sigma_{color=red}(B_1)\right)$

等值连接，条件：sid

sid	sname	rating	age	bid	day
22	dustin	7	45	101	10/10/96
58	rusty	10	35	103	11/12/96

第三步：

$$\pi_{sname,rating,age,bname,color}\left(\left((S_1 \cup S_2)\bowtie_{sid} R_1\right)\bowtie_{bid}\sigma_{color=red}(B_1)\right)$$

选择，条件 color=red

bid	bname	color
101	tiger	red

第四步：

$$\pi_{sname,rating,age,bname,color}\left(\left((S_1 \cup S_2)\bowtie_{sid} R_1\right)\bowtie_{bid}\sigma_{color=red}(B_1)\right)$$

等值连接 条件：bid

sid	sname	rating	age	bid	day	bname	color
22	dustin	7	45	101	10/10/96	tiger	red

第五步：

投影 保留字段：5 个

~~sid~~	sname	rating	age	~~bid~~	~~day~~	bname	color
~~22~~	dustin	7	45	~~101~~	~~10/10/96~~	tiger	red

最终答案

sname	rating	age	bname	color
dustin	7	45	tiger	red

Q7: $\pi_{sid}(S_2) - \pi_{sid}\left(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2)\right)$

第一步：

$$\pi_{sid}(S_2) - \pi_{sid}\left(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2)\right)$$

$\pi_{sid}(\pi_{sid,sname}(S_2))$ $\quad$ $\pi_{sname}(\sigma_{rating>7}(S_1))$

结果：

sid
28
31
44
58

×

sname
Lubber
rusty

等于

sid	sname
28	lubber
28	rusty
31	lubber
31	rusty
44	lubber
44	rusty
58	lubber
58	rusty

第二步：

$$\pi_{sid}(S_2) - \pi_{sid}\left(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2)\right)$$

sid	sname
28	lubber
28	rusty
~~31~~	~~lubber~~
31	rusty
44	lubber
44	rusty
58	lubber
~~58~~	~~rusty~~

减　去

sid	sname
28	yuppy
31	lubber
44	guppy
58	rusty

等于

sid	~~sname~~
28	~~lubber~~
~~28~~	~~rusty~~
31	~~rusty~~
~~44~~	~~lubber~~
44	~~rusty~~
58	~~lubber~~

第三步：

$$\pi_{sid}(S_2) - \pi_{sid}\left(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2)\right)$$

sid
28
31
44
58

减去

sid
28
31
44
58

= NULL

Q8：Conver the query of Q1 to DRC formula and TRC formula.

Q1: $\boldsymbol{\pi}_{\boldsymbol{sid,sname,age}}\left(\boldsymbol{\sigma}_{rating<10}(\boldsymbol{S}_2)\right)$

域关系演算 DRC

$\{ < \boldsymbol{sid, sname, age} > | < \boldsymbol{sid, sname, age} > \in \boldsymbol{S_2} \wedge \boldsymbol{rating} < \boldsymbol{10}\}$

元组关系演算 TRC

$\{\boldsymbol{t}[\ \boldsymbol{sid, sname, age}\ ] \mid \boldsymbol{t} \in \boldsymbol{S_2} \wedge \boldsymbol{t}[\boldsymbol{rating}] < \boldsymbol{10}\}$

Q9：Conver the query of Q2 to DRC formula and TRC formula.

Q2: $\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

域关系演算 DRC

$$\{< sname, rating > | \exists S_1.sid, sname, rating, age, R_1.sid, bid, day$$
$$(< S_1.sid, sname, rating, age > \in S_1 \wedge < R_1.sid, bid, day > \in R_1 \wedge (S_1.sid > R_1.sid))\}$$

元组关系演算 TRC

$\{t[sname, rating] | t = < t^s, t^r > \wedge t^s \in S_1 \wedge t^r \in R_1 \wedge (t[S_1.sid] > t[R_1.sid])\}$

（笛卡尔积）另一种写法供参考

$$R(A) \times S(B) = \{ t \mid \exists(u \in R)\exists(s \in S)(t[A] = u[A] \wedge t[B] = s[B])\}$$

（笛卡尔积）另二种写法供参考

$$R3 = \{t \mid (\exists u)(S(t) \wedge R(u) \wedge t[3] < u[2]\}\}$$

Q10：Conver the query of Q3 to DRC formula and TRC formula.

Q3: $\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

域关系演算 DRC

$$\{< sid, bname > | \exists\, sid, R_1.bid, day, B_1.bid, bname, color(< sid, R_1.bid, day > \in R_1 \wedge < B_1.bid, bname, color > \in B_1 \wedge (R_1.bid = B_1.bid))\}$$

元组关系演算 TRC

$$\{t[\, sid, bname\,] | t =< t^r, t^b > \wedge t^b \in B_1 \wedge t^r \in R_1 \wedge t[R_1.bid] = t[B_1.bid]\}$$

Q11: What’s data model? What’s data schema?

Data model is a collection of concepts and definitions for describing data.

（数据模型是用于描述数据的概念和定义的集合，比如字段）

Data schema is a description of a particular collection of data, using a given data model.

（数据模式是使用给定数据模型对特定数据集合的描述）

姓名	学号	性别	出生年份	籍贯	系别	入学时间

（a）

王彤	0709135	女	1989	江苏	计算机	2007

（b）

图 1-3　数据模式及其实例

Q12: Students(sid, name, , ID number cid)

Courses(cid, cname)

List all super keys, candidate keys, primary keys, foreign keys of table Students.

（1）super keys(超键)

{sid}, {ID number},

{sid, name}, {sid, ID number}, {sid, cid},{ID number, name}, {ID number, cid}, {sid, name, ID number}, {sid, name, cid}, {sid, ID number, cid}, {name, ID number, cid}

{sid, name, ID number, cid}

（2）candidate keys(候补键：)

{sid}, {ID number} 候补键 学号 身份证号都可以唯一标识学生

（3）primary keys(主键:只有一个)

{sid}

（4）foreign keys:(外键)

{cid}

Q13: What are the relational operations on the relational data model? What are their differences and similarities?

Relational calculus vs Relational algebra

(1) Differences:

Relational Algebra needs to specify the order of operations; while relational calculus only needs to indicate the logic condition the result must be fulfilled.

(2) Similiarities:

They are equivalent in terms of expression. SQL language can express any query that is expressible in relational algebra /calculus.